

Step 1

KNIME Analytics Platform

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Local space

Nodes

Example Workflows

KNIME_project_Activity1

CSV Reader

File Mode: File Files in folders

Source: C:\Users\Jojo\Downloads\student_performance_knime...

File Format: Skip first lines of file: 0 Comment line character: #

Discard Apply Apply and Execute

► 1: File Table

Rows: 30 | Columns: 6 % Sorting Filter

#	RowID	student_id	attendance	quiz_score	assignment_score	exam_score	risk_status
1	Row0	S001	95	90	92	88	Not At Risk
2	Row1	S002	88	84	86	82	Not At Risk
3	Row2	S003	76	72	75	70	Not At Risk
4	Row3	S004	62	58	60	55	At Risk

Step 2

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Example Workflows

KNIME_project_Activity1

Column Filter

Column filter: Manual Pattern Type

Excludes: student_id

Includes: attendance, quiz_score, assignment_score, exam_score, risk_status

Discard Apply Apply and Execute

► 1: Filtered table

Rows: 30 | Columns: 5 % Sorting Filter

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status
1	Row0	95	90	92	88	Not At Risk
2	Row1	88	84	86	82	Not At Risk
3	Row2	76	72	75	70	Not At Risk
4	Row3	62	58	60	55	At Risk

Step 3

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Explorer

KAI

Monitor

Workflow diagram showing data processing steps: CSV Reader, Column Filter, Normalizer, Table Partitioner, Logistic Regression Learner, Random Forest Learner, Decision Tree Learner, and Scorer.

Normalizer

Number columns: Manual Pattern Type

Search

Excludes: No columns in this list.

Includes: attendance, quiz_score, assignment_score, exam_score, Any unknown column

Discard Apply Apply and Execute

1: Normalized table 2: Normalize Model Flow Variables

Rows: 30 | Columns: 5 % Sorting Filter

#	RowID	attendance (# Number (Float))	quiz_score (# Number (Float))	assignment_score (# Number (Float))	exam_score (# Number (Float))	risk_status (# String)
1	Row0	0.98	0.92	0.979	0.868	Not At Risk
2	Row1	0.837	0.8	0.854	0.755	Not At Risk
3	Row2	0.592	0.56	0.625	0.528	Not At Risk
4	Row3	0.306	0.28	0.312	0.245	At Risk

Step 4

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Example Workflows

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Explorer

KAI

Monitor

Workflow diagram showing data processing steps: CSV Reader, Column Filter, Normalizer, Table Partitioner, Logistic Regression Learner, Random Forest Learner, Decision Tree Learner, and Scorer.

Table Partitioner

First partition type: Relative (%) Absolute

Relative size: 70

Sampling strategy: Random Stratified Linear First rows

Group column: risk_status

Fixed random seed: 1678807467440

Discard Apply Apply and Execute

1: First partition 2: Second partition Flow Variables

Rows: 21 | Columns: 5 % Sorting Filter

#	RowID	attendance (# Number (Float))	quiz_score (# Number (Float))	assignment_score (# Number (Float))	exam_score (# Number (Float))	risk_status (# String)
1	Row2	0.592	0.56	0.625	0.528	Not At Risk
2	Row3	0.306	0.28	0.312	0.245	At Risk
3	Row4	0.163	0.08	0.146	0.151	At Risk
4	Row5	0.898	0.86	0.917	0.811	Not At Risk

Step 5 - Decision Tree

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Example Workflows

KNIME_project_Activity1

Decision Tree Learner

Class column: risk_status

Quality measure: Gini index Gain ratio

Pruning method: No pruning MDL

Reduced error pruning

Minimum number of records per node: 2

Number of records to store for view: 2

Discard Apply Apply and Execute

1: Decision Tree Model

The data at the output port is not supported by any modern viewer.

Open port view

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Example Workflows

KNIME_project_Activity1

Decision Tree Predictor

Number of patterns for hitting: 10000

Custom prediction column name

Append columns with normalized class distribution

Probability column suffix

Discard Apply Apply and Execute

1: Classified Data

Rows: 9 | Columns: 8

#	RowID	attendance	quiz_score	assignment_s...	exam_score	risk_status	P (risk_status...	P (risk_status...	Prediction (ris...
1	Row0	0.98	0.92	0.979	0.868	Not At Risk	1	0	Not At Risk
2	Row1	0.837	0.8	0.854	0.755	Not At Risk	1	0	Not At Risk
3	Row8	0	0.12	0	0.113	At Risk	0	1	At Risk
4	Row10	0.776	0.74	0.708	0.774	Not At Risk	1	0	Not At Risk

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Example Workflows

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Explorer

KAI

Monitor

Workflow diagram showing a data pipeline: CSV Reader -> Column Filter -> Normalizer -> Table Partitioner -> Logistic Regression Learner -> Logistic Regression Predictor -> Scorer. A Random Forest Learner is also shown as an alternative model.

Scorer

First column: risk_status

Second column: Prediction (risk_status)

Sorting strategy: Insertion order

Missing values: Ignore

Discard Apply Apply and Execute

1: Confusion matrix 2: Accuracy statistics Flow Variables

Rows: 2 Columns: 2 Sorting Filter

#	RowID	Not At Risk Number (Integer)	At Risk Number (Integer)
1	Not A...	5	0
2	At Risk	0	4

Step 6 - Logistic Regression

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Example Workflows

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Nodes

Explorer

KAI

Monitor

Workflow diagram showing a data pipeline: CSV Reader -> Column Filter -> Normalizer -> Table Partitioner -> Logistic Regression Learner -> Logistic Regression Predictor -> Scorer. A Random Forest Learner is also shown as an alternative model.

Logistic Regression Learner

Target column: risk_status

Reference category: Not At Risk

Solver: Stochastic average gradient

Feature selection: Manual

Discard Apply Apply and Execute

1: Model for Predictor 2: Coefficients and Statistics 3: Model and Learning Properties Flow Variables

The data at the output port is not supported by any modern viewer.

Open port view

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Example Workflows

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Logistic Regression Predictor

Custom prediction column name

Append columns with normalized class distribution

Discard Apply Apply and Execute

► 1: Predicted data

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status	Prediction (risk_stat...
1	Row0	0.98	0.92	0.979	0.866	Not At Risk	Not At Risk
2	Row1	0.837	0.8	0.854	0.755	Not At Risk	Not At Risk
3	Row8	0	0.12	0	0.113	At Risk	At Risk
4	Row10	0.776	0.74	0.708	0.774	Not At Risk	Not At Risk

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Example Workflows

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Scorer

First column

risk_status

Second column

Prediction (risk_status)

Sorting strategy

Insertion order

Reverse order

Use name prefix

Missing values

Ignore Fail

Discard Apply Apply and Execute

► 1: Confusion matrix

#	RowID	Not At Risk	At Risk
1	Not A...	5	0
2	At Risk	0	4

Step 7 - Random Forest

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Example Workflows

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Random Forest Learner

Attribute Selection

Target column

risk_status

Training attributes

Use column Use fingerprint

Attribute selection

Manual Pattern

Search

Excludes

Includes

attendance

Discard Apply Apply and Execute

1: Out-of-bag Predictions 2: Attribute Statistics 3: Random Forest Model Flow Variables

Rows: 21 | Columns: 10

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status	P (risk_status)	P (risk_status)	risk_status	risk_status	mod
1	Row2	0.592	0.56	0.625	0.528	Not At Risk					
2	Row3	0.306	0.28	0.312	0.245	At Risk					
3	Row4	0.163	0.08	0.146	0.151	At Risk					
4	Row5	0.898	0.86	0.917	0.811	Not At Risk					

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Example Workflows

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Random Forest Predictor

Change prediction column name

Prediction column name

Prediction (risk_status)

Append overall prediction confidence

Append individual class probabilities

Suffix for probability columns

Use soft voting

Discard Apply Apply and Execute

1: Prediction output Flow Variables

Rows: 9 | Columns: 7

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status	Prediction (risk_status)	Prediction (risk_status)
1	Row0	0.98	0.92	0.979	0.868	Not At Risk	Not At Risk	1
2	Row1	0.837	0.8	0.854	0.755	Not At Risk	Not At Risk	1
3	Row8	0	0.12	0	0.113	At Risk	At Risk	1
4	Row10	0.776	0.74	0.708	0.774	Not At Risk	Not At Risk	1

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Example Workflows

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KAI

Monitor

Scorer

First column: risk_status

Second column: Prediction (risk_status)

Sorting strategy: Insertion order

Missing values: Ignore

Buttons: Discard, Apply, Apply and Execute

1: Confusion matrix

2: Accuracy statistics

Flow Variables

Rows: 2 | Columns: 2

Sorting

Filter

Table

Statistics

Search

#	RowID	Not At Risk	At Risk
1	Not A...	5	0
2	At Risk	0	4